OriginChain: Blockchain-Based Quality Assurance, Authentication and Supply-Chain Traceability for Luxury Cosmetics

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF COMPUTER ENGINEERING

by

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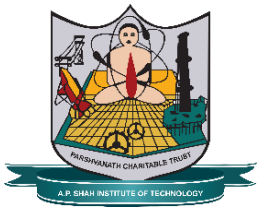


Department of Computer Engineering

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

University of Mumbai

2026-2027



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CERTIFICATE

This is to certify that the project entitled "OriginChain: Blockchain-Based Quality Assurance, Authentication and Supply-Chain Traceability for Luxury Cosmetics" is a bonafide work of **Durva Waghchaure(23102152), Aryan Singh(23102154), Purvika Thombare(23102097) and Aradhya Thakur(23102110)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Engineering**.

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Prof. Pranali Patil

Guide

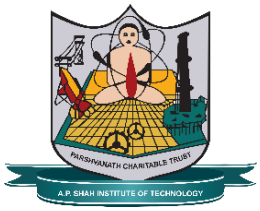
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Project Report Approval for B.E.

This project report entitled ***OriginChain: Blockchain-Based Quality Assurance, Authentication and Supply-Chain Traceability for Luxury Cosmetics*** by *Durva Waghchaure, Aryan Singh, Purvika Thombare* and *Aradhya Thakur* is approved for the degree of *Bachelor of Engineering* in *Computer Engineering*, *2026-27*.

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Counterfeit and mishandled luxury cosmetics are difficult for a buyer to detect: a bottle may look perfect while its contents, batch history or storage conditions differ from what the label claims. Static barcodes and printed seals are easy to copy, and a conventional database can be edited by anyone with enough privilege. This project, OriginChain, proposes and demonstrates a blockchain-based system that combines stage-by-stage quality assurance with tamper-evident product authentication and traceability for luxury cosmetics.

The implemented demonstration follows one fictional flagship product, the *Aurelia Prestige Renewal Serum*, from raw-material receipt through manufacturing, distribution and retail to consumer verification. Each actor (supplier, manufacturer, distributor, retailer) must pass a defined set of quality checks, with supporting PDF/PNG/JPEG evidence, before custody can move on. Evidence files are hashed with Keccak-256; compact fingerprints of the product, of each approved stage and of each evidence file are committed to a Solidity smart contract running on a local Hardhat Ethereum Virtual Machine. Full operational records stay off-chain in SQLite, so the ledger remains small and private. The backend is built with FastAPI and Web3.py; users authenticate with Argon2-hashed passwords and short-lived JSON Web Tokens, and role-based access control is enforced on the server and again by the contract's owner checks.

A consumer can enter a product code and receive a redacted, read-only verification result that reports whether the product is *Authentic* or *Suspicious*, whether every stage was approved, and which evidence the manufacturer chose to make public. A built-in tamper demonstration shows that editing a database record is detected because the recomputed metadata hash no longer matches the on-chain value. Quality gates are mapped to metadata of real standards (the Cosmetics Rules 2020 and selected BIS and ISO cosmetics standards) purely as traceability references; the project makes no certification or compliance claim. The current build passes 51 backend tests and 17 smart-contract tests.

Contribution to Sustainable Development Goals (SDGs). OriginChain supports SDG 12 (Responsible Consumption and Production) by making product origin and handling transparent and by discouraging counterfeit goods; SDG 9 (Industry, Innovation and Infrastructure) through an innovative, auditable digital infrastructure for supply chains; SDG 3 (Good Health and Well-being) by reducing the chance that unsafe or substandard cosmetics reach users; and SDG 16

(Peace, Justice and Strong Institutions) through accountable, attributable and tamper-evident records.

Layout of the report

The first chapter introduces the problem and the project. The second reviews the literature on blockchain traceability, anti-counterfeiting and cosmetics quality standards. The third states the limitations of existing systems and the research gap, and the fourth gives the problem statement, objectives and scope. The fifth presents the proposed system with its architecture and UML diagrams, algorithms, design details and methodology. The sixth describes the experimental setup, the seventh the project plan, the eighth the expected outcome and the ninth the projected cost analysis, followed by the references.

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ABBREVIATIONS

Abbreviation	Expansion
ABI	Application Binary Interface
API	Application Programming Interface
BIS	Bureau of Indian Standards
CDSCO	Central Drugs Standard Control Organisation
COA	Certificate of Analysis
CSS	Cascading Style Sheets
ER	Entity-Relationship
EVM	Ethereum Virtual Machine
GMP	Good Manufacturing Practice
HTML	HyperText Markup Language
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
IPFS	InterPlanetary File System
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
JWT	JSON Web Token
NFC	Near Field Communication
QC	Quality Control
RBAC	Role-Based Access Control
REST	Representational State Transfer
RFID	Radio-Frequency Identification
SDG	Sustainable Development Goal
SQL	Structured Query Language
SRS	Software Requirements Specification
UML	Unified Modeling Language

Chapter 1

Introduction

1.1 Background and Motivation

Counterfeiting is a large and growing problem in global trade. An OECD and EUIPO study estimated that counterfeit and pirated goods made up roughly 3.3 percent of world trade in 2016 [1]. Luxury cosmetics are particularly exposed: they carry a high price, are easy to refill or re-label, and a buyer cannot judge a sealed bottle's authenticity by its appearance. Unlike a counterfeit handbag, a counterfeit serum is applied to the skin, so contamination, wrong ingredients or poor storage can cause irritation or infection. Consumers, brands and regulators therefore need a trustworthy way to confirm that a product is genuine and was handled correctly.

Printed barcodes, holograms and static QR codes can be duplicated, and centralised databases depend on a single operator, so a privileged insider can silently alter a record. Blockchain offers a different trust model: an append-only ledger whose blocks are linked by cryptographic hashes, so altering an old entry invalidates every later block [2], [3]. Smart contracts on platforms such as Ethereum enforce rules without a trusted intermediary [4], [5], making blockchain a strong foundation for traceability, provided that large and sensitive data stay off the chain [6]. We therefore asked what a luxury-cosmetics supply chain really needs: proof that each party passed meaningful quality checks, tamper-proof evidence, controlled access for each role, and a consumer view that builds trust without leaking internal information.

Sustainable Development Goals. OriginChain supports SDG 3 (Good Health and Well-being) by reducing the chance that counterfeit cosmetics reach consumers, SDG 9 (Industry, Innovation and Infrastructure) through auditable blockchain-based infrastructure, SDG 12 (Responsible Consumption and Production) through transparent product origin, and SDG 16 (Peace, Justice and Strong Institutions) through tamper-evident records.
Program Outcomes. PO1 and PO2 are met through engineering fundamentals and the literature survey, PO3 through the system design, PO5 through tools such as FastAPI, Solidity and Hardhat, PO6 and PO7 through consumer safety and societal impact, PO9 and PO10 through teamwork, this report and the demonstration, and PO11 through project planning and cost analysis.
Program Specific Outcomes. PSO1 is met by building a complete working application, PSO2 by following standard software practices with 68 automated tests, and PSO3 by using modern blockchain technology that can lead to entrepreneurship, higher studies or industry work.

1.2 Overview of OriginChain

OriginChain is a working, end-to-end demonstration that tracks one fictional luxury anti-aging serum, the *Aurelia Prestige Renewal Serum* (product code OC-LUX-SERUM-0001, batch APR-2026-001), from raw-material receipt to the retail shelf. The product, the companies and all inspection results are fictional demonstration data; only the standards metadata that the quality checks refer to is real. The system is built around four ideas.

- **Stage-gated quality assurance.** The supply chain is modelled as four operational stages: raw-material supplier, manufacturer, distributor and retailer. Each stage has a defined list of quality checks (24 in total: 5 supplier, 8 manufacturer, 5 distributor and 6 retail). A stage can be approved, and custody can be transferred, only when every mandatory check passes and every required piece of evidence exists and is intact.
- **Evidence with integrity and trust.** Certificates, laboratory reports and inspection photographs are uploaded as PDF, PNG or JPEG files, hashed with Keccak-256, and re-verified on demand. For critical reports (certificate of analysis, microbiology and preservation efficacy) the issuer must also be currently trusted in a local registry of fictional issuers.
- **Details off-chain, fingerprints on-chain.** Full records and files remain in SQLite and on disk. The Solidity contract stores only 32-byte hashes, owner addresses, status codes and events, which keeps the ledger compact and private while still making tampering detectable.
- **Consumer verification.** Anyone can enter a product code and receive a redacted result: Authentic or Suspicious, the four-stage quality journey, provenance by role, and any evidence explicitly marked public. Internal notes, files and identifiers are never exposed.

Behind these ideas sit a FastAPI backend with role-based access control, an audit log of every meaningful action, a single-page web interface that adapts to the signed-in role, and a set of automated tests. A built-in tamper demonstration lets an administrator edit a stored record and then shows the consumer view turning from Authentic to Suspicious, which makes the value of the on-chain fingerprint visible in a presentation.

1.3 Why Luxury Cosmetics

Cosmetics are a good test case for a quality-aware traceability system for three reasons. First, they combine high value with safety sensitivity, so both authenticity and handling matter. Second, a finished cosmetic depends on several upstream parties (ingredient suppliers, a contract or in-house manufacturer, distributors and retailers), so no single party can vouch for the whole history. Third, there is a rich body of rules and standards that explain *why* particular evidence is requested. In India, the Cosmetics Rules, 2020 govern manufacture, labelling and traceability information [7]; ISO 22716 gives guidelines for good manufacturing practice [8]; and a group of ISO microbiology standards covers microbial limits, risk assessment and the evaluation of antimicrobial protection [9], [10], [11]. OriginChain records metadata about these sources and maps selected quality gates to them so that a reviewer can see why each piece of evidence is requested. The mapping is only a reason to request or inspect evidence; it is never a claim that the product complies with, or is certified under, any standard.

1.4 Technologies Used

The demonstration deliberately uses a small, widely available technology stack so that it can run on a single laptop. The backend is written in Python 3.13 with FastAPI and communicates with the blockchain through Web3.py. Operational data is kept in SQLite. The smart contract is written in Solidity (version 0.8.28 or later) and is compiled, tested and deployed with Hardhat 3 against a local Ethereum Virtual Machine. The user interface is plain HTML, CSS and JavaScript served by FastAPI. Passwords are stored as Argon2 hashes, a memory-hard design that resists offline guessing [12], and sessions use signed, short-lived JSON Web Tokens [13]. Evidence files and product metadata are fingerprinted with Keccak-256, the hash function used by Ethereum.

1.5 Significant Contributions

1. A hybrid on-chain/off-chain architecture for cosmetics traceability in which only compact commitments (product metadata hash, approved-stage snapshot hash and evidence hashes) are written to the ledger.
2. A stage-gated quality model with 24 checks, required-evidence rules, live re-evaluation at every transfer, and automatic invalidation of an approval when a result or evidence is edited.
3. A two-part evidence model that separates file integrity (do the stored bytes still match the recorded hash?) from issuer verification (was the report accepted from a currently trusted issuer?), with frozen issuer snapshots so that later revocation does not silently rewrite history.
4. Layered authorization: server-side role and stage rules in FastAPI, plus independent current-owner checks in the smart contract.
5. A privacy-preserving consumer verification view with explicit evidence visibility controls, and a tamper demonstration that makes integrity failures visible.
6. A traceability layer that links quality gates to real standards metadata without copying copyrighted text and without claiming certification.

1.6 Organization of the Report

Related work is surveyed in chapter 2. Limitations of existing systems are identified in chapter 3, and the problem, objectives and scope are stated in chapter 4. The proposed system, including architecture, UML diagrams, algorithms and methodology, is described in chapter 5. Chapter-wise, the experimental setup and evaluation parameters follow in chapter 6, the project plan in chapter 7, the expected outcome in chapter 8 and the projected cost in chapter 9. The report ends with the references.

Chapter 2

Literature Survey

2.1 Overview

This chapter reviews published work that bears on the project. It begins with the blockchain and smart-contract foundations, moves to blockchain-based supply-chain traceability and anti-counterfeiting, looks at deployments in food and pharmaceutical chains that resemble the cosmetics case, and finishes with the security building blocks and the cosmetics standards on which the quality model rests. Each group of papers is appraised for what it contributes and for what it leaves open for this project.

2.2 Blockchain and Smart-Contract Foundations

The idea of a blockchain was introduced by Nakamoto, who described a peer-to-peer electronic cash system in which transactions are grouped into blocks that are chained by cryptographic hashes and secured by proof of work, removing the need for a trusted third party to prevent double spending [2]. The same chain-of-hashes principle is what makes tampering with historical records detectable in OriginChain: a stored fingerprint cannot be changed without leaving evidence.

Wood generalised the model in the Ethereum Yellow Paper, which formalises a transaction-based state machine and an Ethereum Virtual Machine that executes programs, called smart contracts, identically on every node [4]. Smart contracts allow business rules such as "only the current owner may transfer custody" to be enforced by the network rather than by an operator. OriginChain uses exactly this capability: its contract rejects ownership transfers and commitment writes from any address other than the current owner.

Zheng *et al.* provide a widely cited overview of blockchain architecture, consensus mechanisms and future trends [3], and a companion survey of challenges and opportunities that highlights scalability, privacy leakage, storage growth and the cost of on-chain data [14]. These challenges

explain why OriginChain does not store full records on the chain. Instead it keeps only hashes and status codes on the ledger and the detailed data off-chain, which is consistent with the guidance that on-chain storage should be reserved for what must be tamper-evident.

Christidis and Devetsikiotis examine how blockchains and smart contracts can coordinate devices and parties in the Internet of Things without a central authority, and discuss the autonomy and trust benefits as well as the open issues of cost and performance [5]. Reyna *et al.* review the integration of blockchains with IoT and conclude that scalability, privacy, interoperability and the management of off-chain data are the main obstacles [15]. For a cosmetics chain, where evidence files such as laboratory reports are far too large to store on-chain, their analysis supports the hash-anchoring approach adopted here.

2.3 Blockchain-Based Supply-Chain Traceability

Tian proposed one of the earlier traceability systems that couples RFID tagging with blockchain for the agri-food supply chain in China, using the shared ledger to make information about each stage visible and hard to alter [16]. The work shows how a multi-party chain can obtain a common, trusted record, but it focuses on identification and information sharing and does not define approval gates that stop a product from moving on when quality conditions are not met.

Saberi *et al.* study how blockchain relates to sustainable supply-chain management and classify the barriers to adoption into intra-organisational, inter-organisational, system-related and external groups [17]. They argue that blockchain can improve transparency, accountability and certification practices, but that technical immaturity and the need for collaboration between partners slow adoption. This framing is useful for OriginChain because the demonstration deliberately models four cooperating organisations and treats organisational roles as first-class concepts.

Pournader *et al.* carried out a systematic review of blockchain applications in supply chains, transport and logistics and identified themes such as traceability, provenance, trust and smart-contract automation, while noting that much of the literature remains conceptual and that empirical, implementation-level studies are scarce [18]. Sunny *et al.* respond to this gap with a traceability overview accompanied by a working demonstration, showing how supply-chain

events can be recorded in smart contracts and queried for transparency [19]. Their demonstration confirms feasibility, yet the quality dimension is limited to recording events rather than verifying that the necessary quality evidence is present before an event is allowed.

Westerkamp *et al.* describe how manufacturing processes can be traced using token compositions on a blockchain, so that a finished good carries a verifiable link to the components and process steps that produced it [20]. This idea of composing a finished product from the records of its inputs is close to OriginChain's linkage of a finished batch to approved raw-material batches, which is enforced by the backend and summarised by a deterministic ingredient-lineage commitment.

Lu and Xu discuss a case study of blockchain-based product traceability from the viewpoint of software architecture, with attention to design decisions such as what should be stored on-chain and what off-chain, and how the system can be adapted to different products [6]. Their emphasis on architecture and adaptability directly informs the layered design adopted in chapter 5 and the decision to model product category, subcategory and type so that other cosmetic products can be added later.

2.4 Anti-Counterfeiting and Ownership Management

Toyoda *et al.* propose a blockchain-based product ownership management system (POMS) aimed at anti-counterfeiting in the *post* supply chain, that is, after a product has left the manufacturer, including resale between owners [21]. A smart contract records ownership transfers so that a buyer can verify that a seller is the legitimate owner and that the product is genuine. The proposal is strong on ownership and on incentives for honest reporting, but it treats authenticity as a property of the ownership chain and does not look at the quality state of the product at each hand-off.

Alzahrani and Bulusu present Block-Supply Chain, an anti-counterfeiting design that combines NFC tags with a blockchain so that a tagged item can be authenticated and its path checked by a smartphone [22]. Tagging hardware gives a physical anchor between the item and its record, which OriginChain does not yet implement; the demonstration instead provides a QR code

endpoint for the verification URL. Physical-tag binding, and its own attack surface, therefore remains an item of future work.

2.5 Food and Pharmaceutical Supply Chains

Kamilaris *et al.* review blockchain in agriculture and food supply chains and report that food traceability, safety and provenance are the leading applications, with most projects still at pilot stage [23]. The food chain resembles the cosmetics chain in that safety and handling conditions are as important as origin. The review also notes the lack of standard data models and the dependence on how trustworthy the data entered into the chain is, a problem sometimes described as "garbage in, garbage out".

Bocek *et al.* present a use case for blockchains in the pharmaceutical supply chain in which sensor readings such as temperature during transport are recorded through smart contracts so that breaches can be detected and the information shared among parties [24]. Pharmaceuticals and cosmetics both involve strict storage conditions and counterfeit risk, and the paper demonstrates that a ledger can bind condition data to a product journey. Like most of the work above, it focuses on automatic data capture and leaves open how documentary evidence and the trustworthiness of the issuing laboratory should be handled.

2.6 Security Building Blocks

Because OriginChain exposes a web interface for operational users, the security of identities and sessions matters as much as the ledger. Biryukov *et al.* introduce Argon2, a memory-hard password-hashing function that raises the cost of large-scale offline guessing attacks on stolen password databases [12]; OriginChain stores only Argon2 hashes of the demo passwords. The JSON Web Token standard defines a compact, signed format for carrying claims between parties [13]; OriginChain issues tokens that expire after 30 minutes and, on every protected request, reloads the user and confirms that the account is still active and still holds the role named in the token. A valid token is therefore necessary but not sufficient.

2.7 Cosmetics Regulation and Standards

A quality-assurance model for cosmetics should be anchored in recognised sources. The Cosmetics Rules, 2020 of India set out requirements for the manufacture and labelling of cosmetics, including information that supports traceability such as the manufacturer, batch and dates [7]. ISO 22716 provides guidelines for good manufacturing practice covering production, control, storage and shipment of cosmetic products [8]. For microbiological quality, ISO 17516 deals with microbiological limits [9], ISO 29621 with risk assessment and the identification of microbiologically low-risk products [10], and ISO 11930 with the evaluation of the antimicrobial protection of a cosmetic product [11]. OriginChain records metadata (authority, title, edition, lifecycle status and official URL) for these and related sources, and maps selected quality gates to them. It does not reproduce standard text, test procedures or numeric limits, and it never presents the mapping as certification.

2.8 Comparative Summary

Table 2.1 summarises the reviewed approaches in terms of domain, technique and the gap that remains from the point of view of this project.

Table 2.1: Summary of related work

Reference	Domain	Technique	Remaining gap for OriginChain
Tian [16]	Agri-food	RFID + blockchain traceability	No approval gates; identification and sharing only
Saberi et al. [17]	Sustainable SCM	Conceptual analysis of barriers	No implementation; no quality workflow
Pournader et al. [18]	Supply chain, logistics	Systematic review	Notes scarcity of implementation-level studies
Sunny et al. [19]	General SCM	Smart-contract traceability demonstration	Records events; does not verify evidence before an event
Westerkamp et al. [20]	Manufacturing	Token compositions for components	Provenance of parts; no evidence or issuer trust model
Lu and Xu [6]	Product traceability	Architecture case study	General design; not tied to cosmetics standards
Toyoda et al. [21]	Anti-counterfeit	Ownership management by smart contract	Ownership only; no stage-wise quality state

Reference	Domain	Technique	Remaining gap for OriginChain
Alzahrani and Bulusu [22]	Anti-counterfeit	NFC tags + blockchain	Needs tag hardware; no quality evidence
Kamilaris et al. [23]	Food	Review of applications	Data trustworthiness left open
Bocek et al. [24]	Pharma	Sensor data via smart contracts	Documentary evidence and issuer trust not handled

2.9 Critical Appraisal

Taken together, the literature establishes that blockchains can make records tamper-evident, that smart contracts can enforce simple rules, and that traceability and anti-counterfeiting are promising applications. Three observations follow from the survey. First, authenticity is usually reduced to identity and ownership; the quality state of a product at each hand-off is rarely a precondition for the next step. Second, most proposals trust whatever data are entered, and few treat evidence documents, their integrity and the credibility of their issuers as explicit, verifiable objects. Third, work that touches cosmetics is limited, and the link from a quality check to a recognised regulation or standard is rarely made traceable. These observations define the research gap that OriginChain addresses and are developed further in chapter 3.

Chapter 3

Limitation of Existing System

The literature in chapter 2 and current practice leave the following research gap.

1. **Copyable or editable records.** Printed codes and holograms can be duplicated, and a central verification database can be altered by a privileged insider without detection.
2. **Authenticity without quality.** Blockchain anti-counterfeiting designs [21], [22] prove identity and ownership but do not require passed quality checks before custody changes.
3. **Unverified inputs.** Submitted data are usually trusted [23]; evidence documents are not hash-bound and issuer credibility is not modelled.
4. **Weak standards linkage.** Quality requirements are rarely traceable to a named, versioned source or edition.
5. **On-chain bloat and thin role separation.** Full records on-chain raise cost and privacy risk [14], and many prototypes use one signer with no independent owner check.
6. **No cosmetics focus.** Surveyed works target food, pharma and general goods; none handles cosmetic evidence such as microbiology and preservation efficacy.

Chapter 4

Problem Statement, Objectives and Scope

4.1 Problem Statement

Luxury cosmetics are vulnerable to counterfeiting and mishandling, yet buyers cannot verify a product's authenticity or whether every party in its supply chain performed the required quality checks. Existing authentication methods are copyable or rely on a single editable database, and existing blockchain proposals rarely tie custody changes to verified quality evidence. A system is needed that makes product identity, stage-wise quality approval and supporting evidence tamper-evident and verifiable by the consumer, without exposing internal data.

4.2 Objectives

1. To record product identity and custody on an append-only ledger that the application database cannot silently rewrite.
2. To enforce stage-by-stage quality gates (supplier, manufacturer, distributor, retailer) so that a product cannot be transferred or sold until the current stage is approved.
3. To tie every approval to evidence (certificates, reports, inspection photographs) whose integrity can be re-checked later using Keccak-256 hashes.
4. To model the trustworthiness of evidence issuers separately from file integrity, preserving the status that applied at submission time.
5. To provide role-based access control and an attributable audit log of all significant actions.
6. To give consumers a read-only, privacy-preserving verification view that reports Authentic or Suspicious.

7. To demonstrate tamper detection: an edited database record must be visibly flagged as Suspicious.
8. To make quality requirements traceable to versioned sources and standards metadata without claiming certification.

4.3 Scope

In scope

- One fictional flagship product (anti-aging facial serum) and one workflow across four operational stages and a public consumer view.
- A local, single-machine demonstration: FastAPI, SQLite, Web3.py, a Solidity contract on a local Hardhat network, and a static web interface.
- Seeded fictional users, evidence issuers, quality templates and product specification, with an India-oriented demonstration quality profile.
- Automated testing of backend logic, access control, quality gates and the smart contract.

Out of scope (future work)

- Production identity management, multi-factor authentication, key custody and secrets management.
- Public blockchain deployment, IPFS or cloud object storage, and PostgreSQL migration.
- AI or machine-learning packaging inspection; no model or confidence score is implemented.

Applicability

The design generalises to other categories (moisturiser, eye cream, cleanser, makeup, fragrance) and, with changes to the quality templates, to other regulated goods in which authenticity and handling must both be proven. The usefulness of the project lies in showing a complete, testable pattern that a brand or auditor could extend.

Chapter 5

Proposed System

5.1 Analysis and Framework

The analysis started from the question of what must be provable to a consumer: that the product is the one the manufacturer registered, that each party passed its checks, that the supporting evidence has not changed, and that none of the stored records were edited afterwards. The framework that answers it rests on four principles.

1. **Details off-chain, fingerprints on-chain.** Large and private data stay in SQLite and on disk. The ledger holds only hashes, owner addresses, status codes and events.
2. **Gate before movement.** Custody cannot move, and a product cannot be approved for sale, until the current stage's mandatory checks and required evidence are satisfied. Gates are re-evaluated live rather than trusted from a stored approval.
3. **Separation of identities.** Application identity (user, role) is separate from blockchain execution identity (a server-held signer). Passwords and tokens never act as blockchain keys, and no private key reaches the browser.
4. **Honest claims.** Standards are referenced as metadata only; "verified for demo" means accepted against local fictional controls and nothing more.

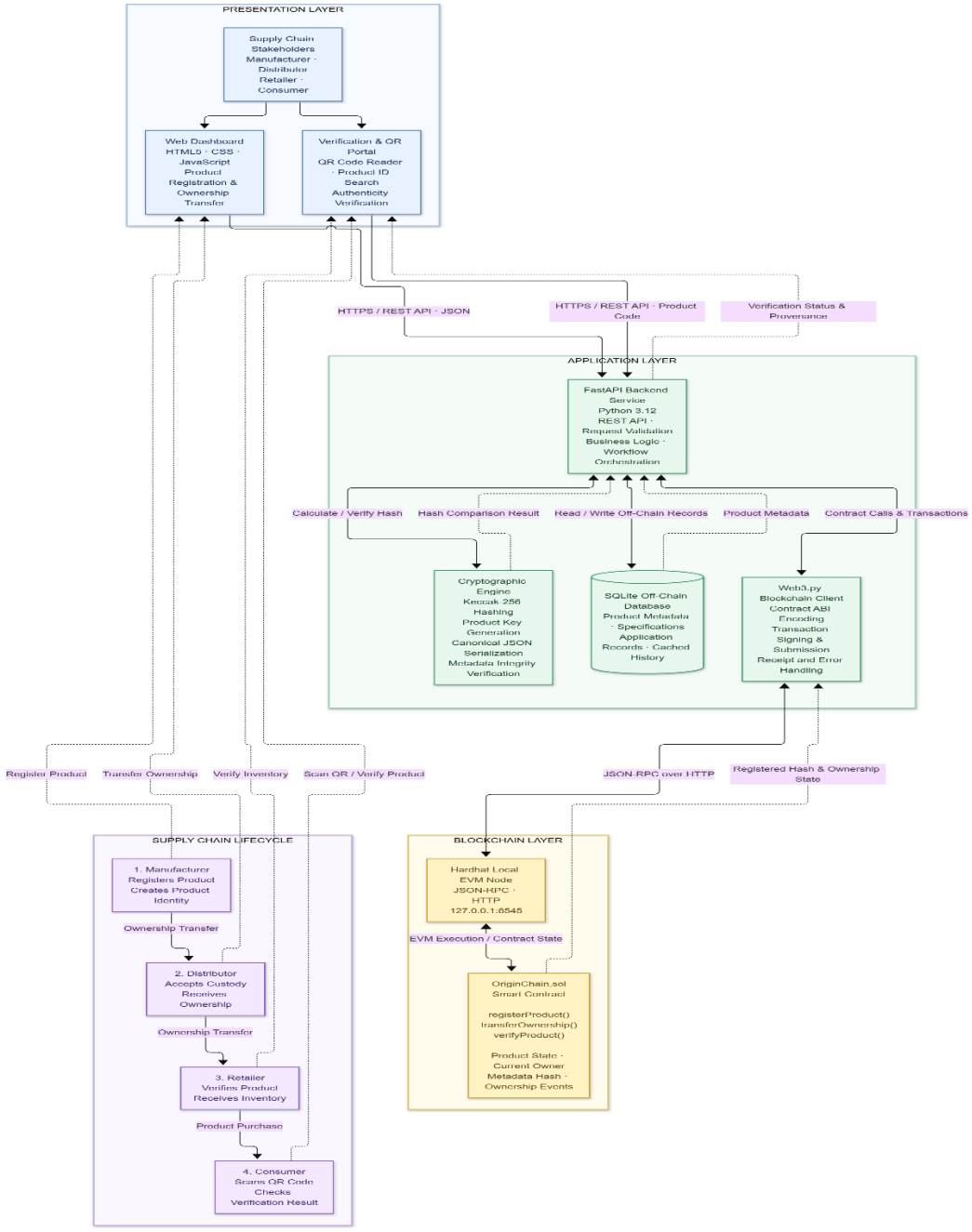
Assumptions and threat model

The demonstration assumes a single trusted host that runs the API, the database and a local blockchain. The threats addressed are post-hoc editing of stored product records, replacement or modification of evidence files, unauthorised role actions, skipping a quality stage, and exposure of internal data to consumers. The threats that remain, including compromise of the host or of the server-held signing keys, dishonest original inspections and production-grade identity attacks, are listed as limitations. The blockchain proves that a record has not changed since it was committed; it cannot prove that the original inspection was honest.

5.2 System Architecture

The system architecture diagram shows how OriginChain's presentation, application, blockchain, and supply chain components work together for product registration, ownership tracking, and authentication.

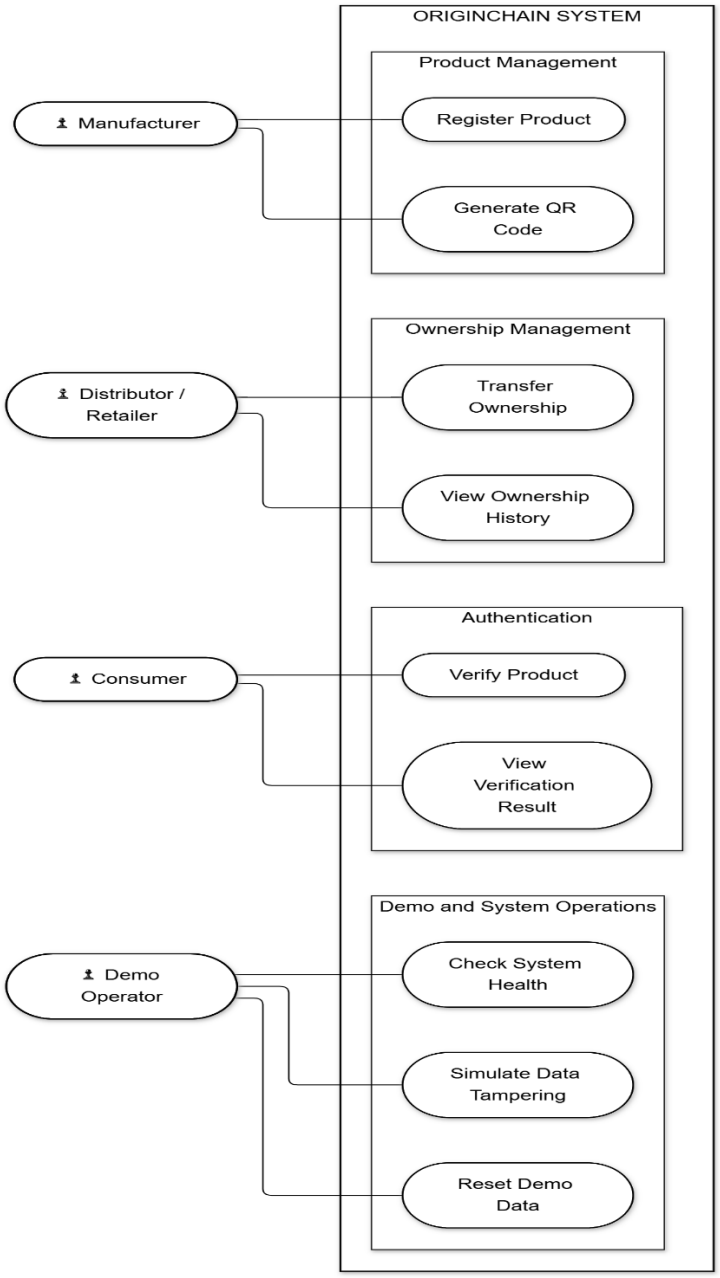
Fig 5.2: Architecture Diagram



5.3 Use Case Diagram:

The use case diagram shows how the Manufacturer, Distributor/Retailer, Consumer, and Demo Operator interact with OriginChain's main functions, including product registration, ownership transfer, QR generation, verification, and demo operations.

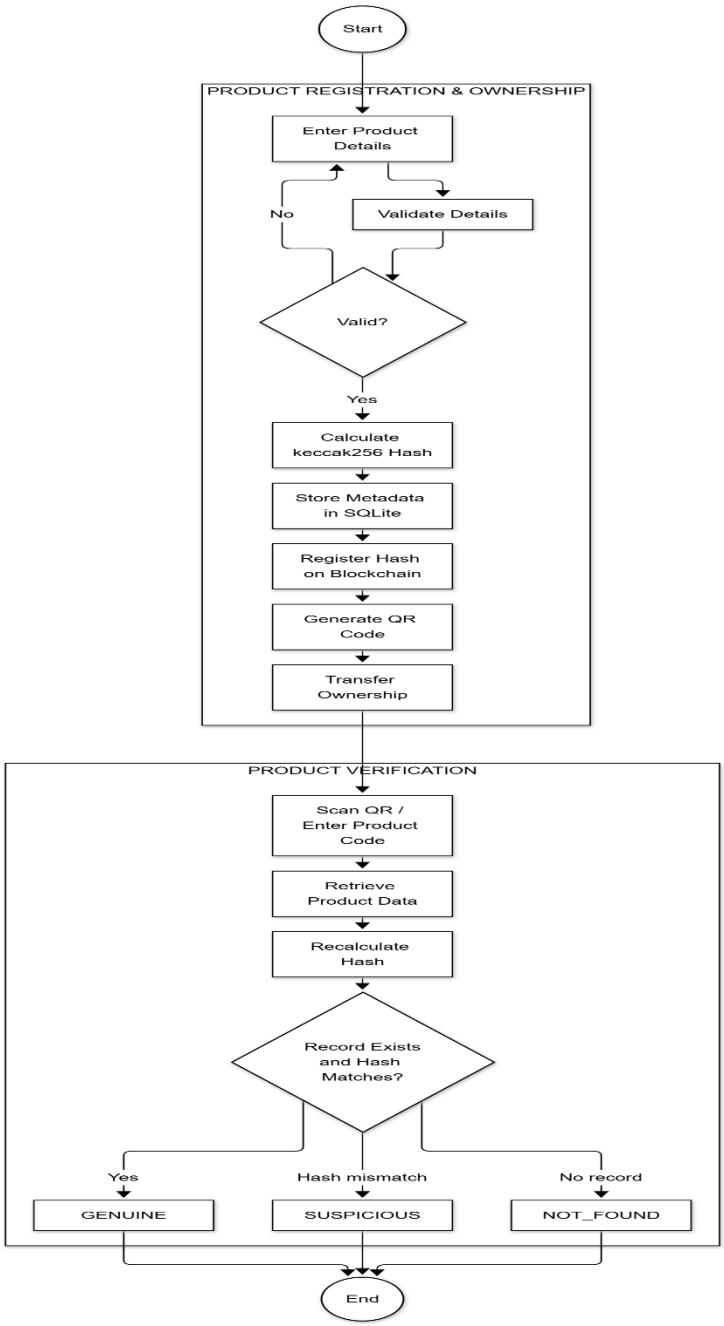
Fig 5.3 Use Case Diagram



5.4 Workflow (Activity) Diagram

The activity workflow diagram shows OriginChain's product registration, ownership transfer, and QR-based verification process. It identifies genuine, suspicious, and unregistered products by comparing metadata hashes.

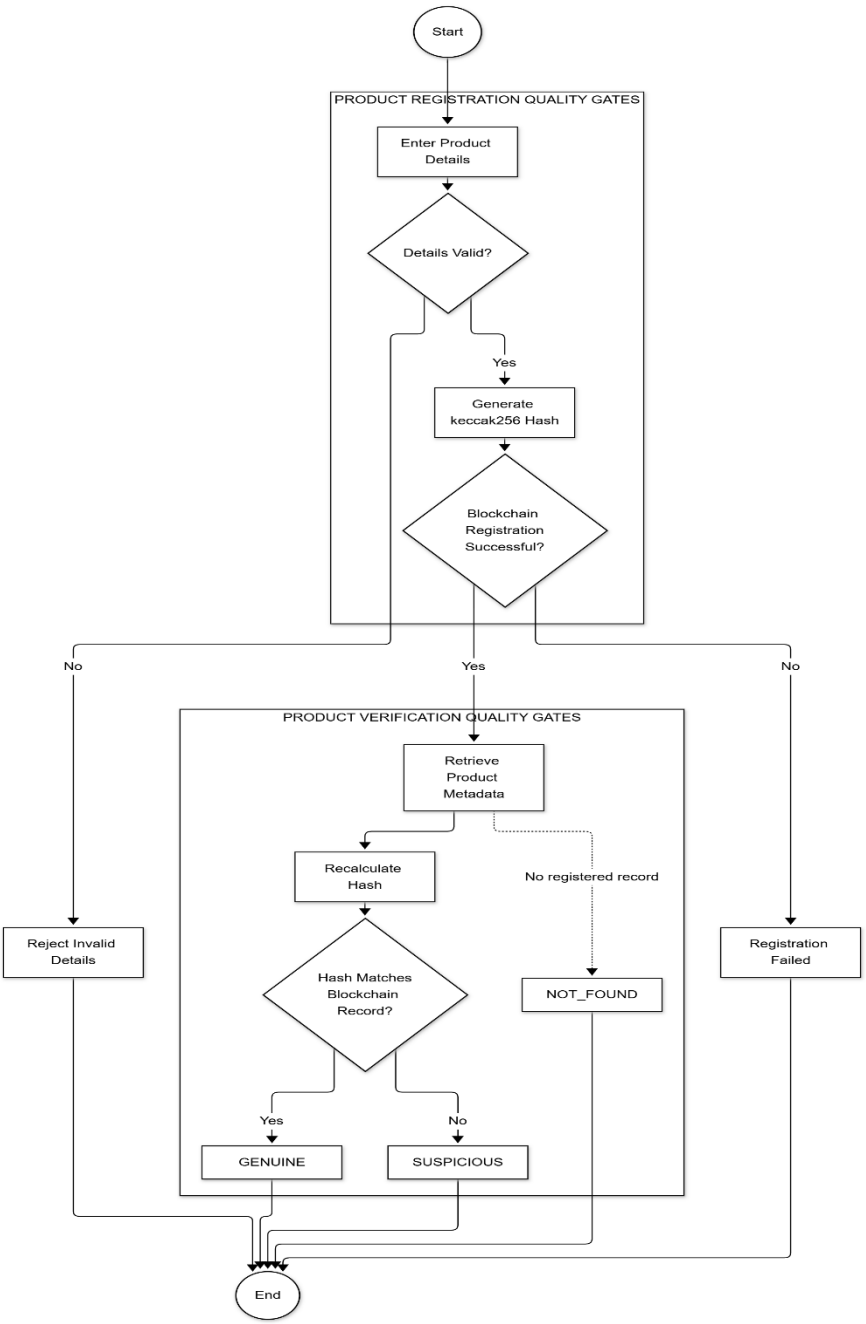
Fig 5.4: Workflow (Activity) Diagram



5.5 Quality-Gate Flow

The quality gate flow diagram shows the validation checks performed during product registration and verification in OriginChain. It ensures valid details, successful blockchain registration, and hash integrity before determining product authenticity.

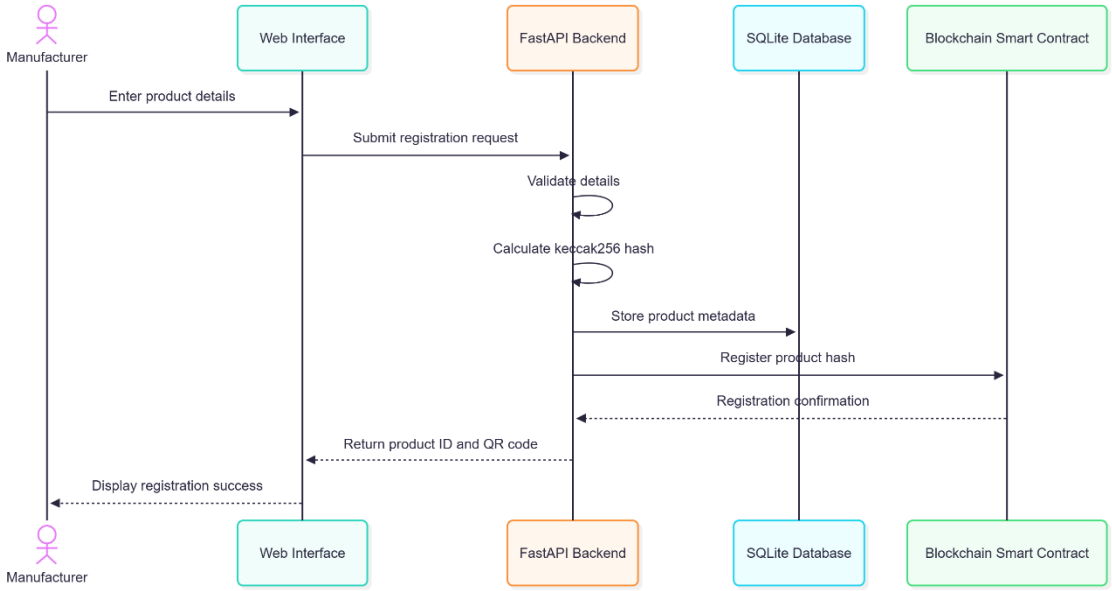
Fig 5.5: Quality-Gate Flow Diagram



5.6 Sequence Diagram

The sequence diagram illustrates the interaction between the manufacturer, web interface, backend, SQLite database, and blockchain during product registration. It shows how product details are validated, stored, hashed, and registered before returning the product ID and QR code.

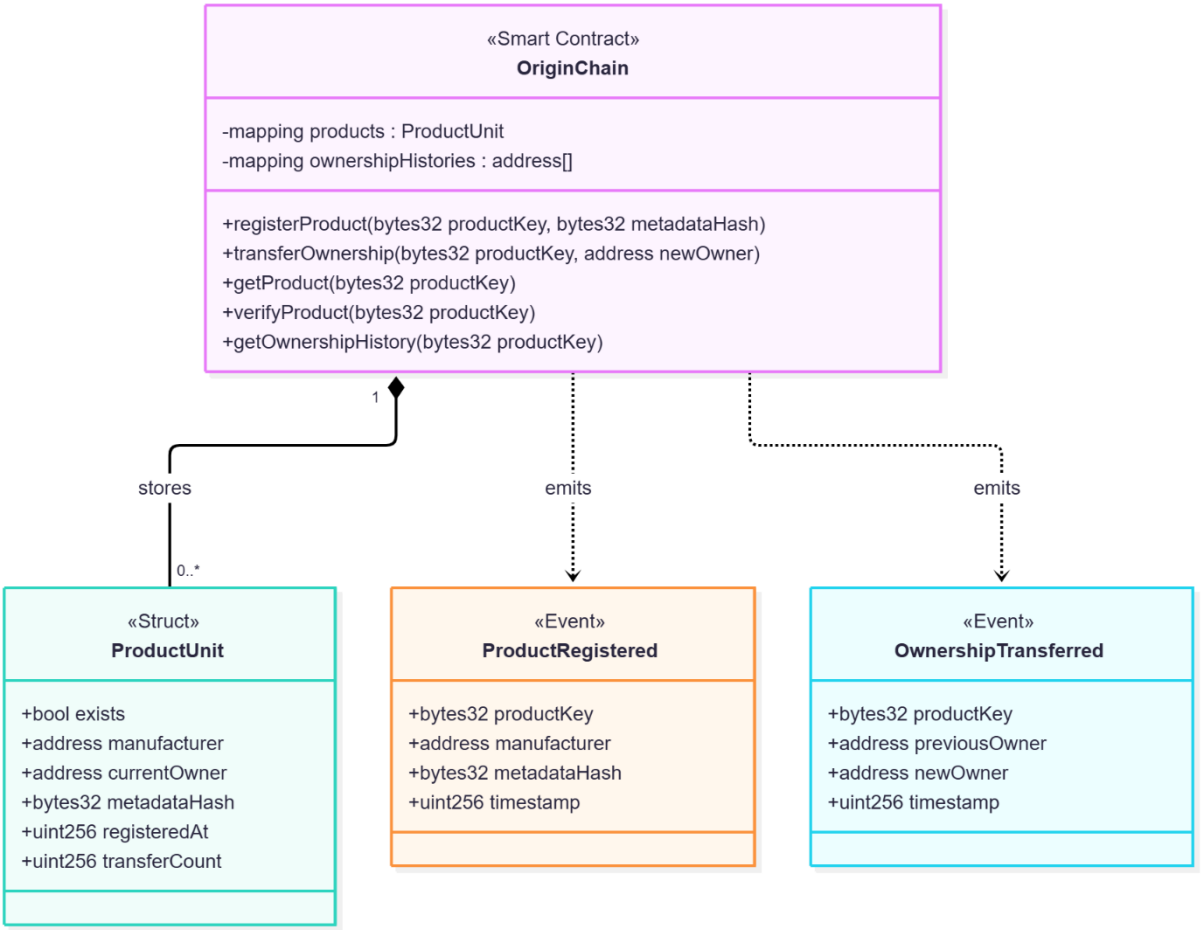
Fig 5.6: Sequence Diagram



5.7 Class Diagram of the Smart Contract

The diagram shows OriginChain's smart contract structure, including product data, functions, and events. It illustrates product registration, ownership transfers, and ownership history management.

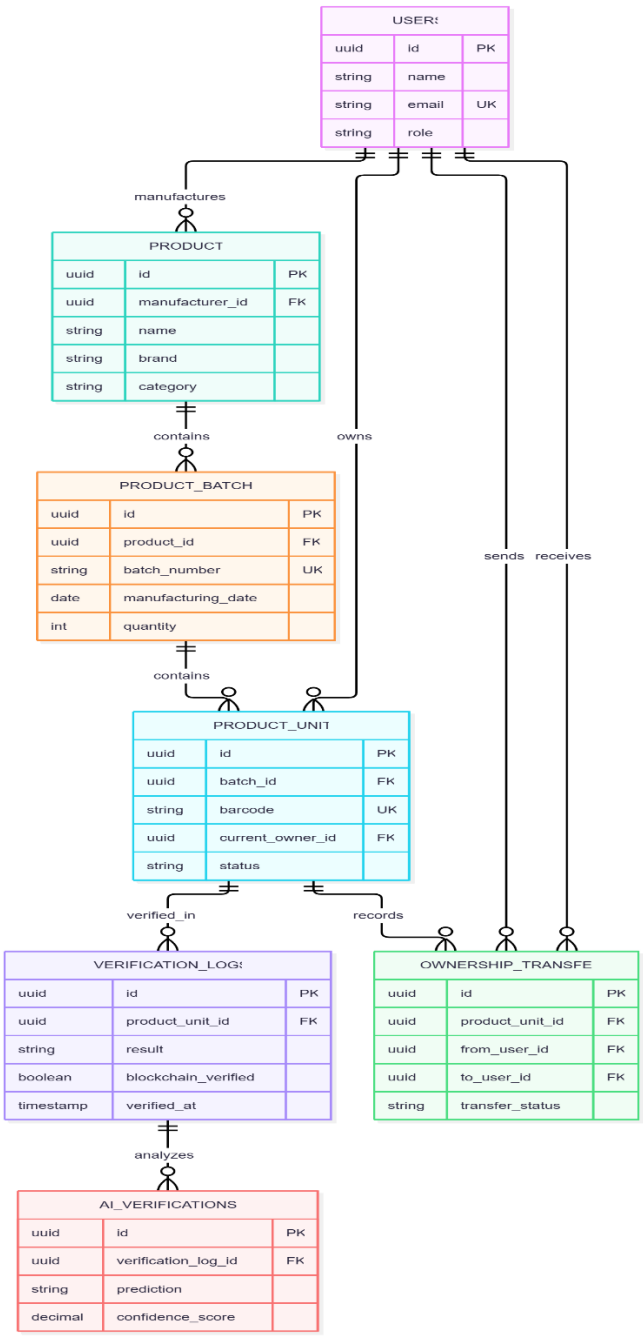
Fig 5.7: Class Diagram of Smart Contract



5.8 Entity-Relationship Diagram

The ER diagram illustrates OriginChain's main entities, including users, products, batches, and product units. It shows how ownership transfers, product verification, and AI-based analysis are connected.

Fig 5.8: Entity-Relationship Diagram



5.9 Algorithms

Algorithm 1: Product metadata hash

1. Build a metadata object with the fields batch_number, brand, description, name and product_code.
2. Serialise it to canonical JSON with keys sorted, no extra whitespace and non-ASCII characters kept.
3. Compute H = Keccak-256 of the canonical text and the product key K = Keccak-256 of the trimmed product code.
4. Call registerProduct(K, H) on the contract; store K, H and the transaction hash in SQLite.

Algorithm 2: Stage approval

1. Load all checks for the stage and the context (raw material or product).
2. If any mandatory check is FAIL, set the stage to HOLD and list the blockers; if any is not yet PASS, keep it PENDING.
3. For each check that requires evidence, confirm that a file of the expected document type exists and that its stored bytes still match the recorded hash.
4. For critical reports, confirm a report number and an issuer accepted as TRUSTED_DEMO at submission.
5. For the manufacturer stage, confirm at least one linked, approved and unexpired raw-material batch.
6. If all conditions hold, build the stage snapshot, commit its hash and each evidence hash on-chain, and mark the stage APPROVED.

Algorithm 3: Consumer verification

1. Look up the product by code; if it is not found, return NOT_FOUND.
2. Recompute the metadata hash from the stored record and read the on-chain hash.
3. If the two differ, return SUSPICIOUS with a hash-mismatch indication.
4. Otherwise return AUTHENTIC together with the redacted stage journey, role-only provenance, final sale status and consumer-visible evidence.

5.10 Design Details

Modules

Table 5.1: Backend modules and responsibilities

Module	Responsibility
main.py	FastAPI application and routes for authentication, records, quality, evidence, transfers, verification, administration and demo support
database.py	SQLite schema (20 tables), repositories and guards that block edits to assigned specifications
domain.py	Roles, statuses, stage definitions and the seeded quality templates (24 checks)
quality_service.py	Evaluation of stage gates, required checks, evidence and issuer trust
auth.py	Argon2 password hashing, 30-minute JWTs and per-request user reload
evidence.py	Upload validation (PDF/PNG/JPEG, up to 5 MiB), generated filenames and byte hashing
metadata_hash.py	Canonical metadata serialisation and Keccak-256 hashing
blockchain_client.py	Web3.py wrapper that maps each role to a server-held signer
presentation_demo.py	Demo auto-fill and full presentation seeding that use the real gates

Roles and access

Table 5.2: Roles and permitted actions

Role	Permitted actions
Supplier	Register raw materials; perform and approve raw-material QC
Manufacturer	Register finished product; manufacturing QC; transfer to Distributor
Distributor	Distribution QC; transfer to Retailer
Retailer	Retail QC and the final sale decision
Admin	Read all stages, audit log, issuer trust and governance records; run demo tools; cannot perform supply-chain actions
Consumer	Public verification only; no account

Data placement

Table 5.3: Where each kind of data is stored

Location	Content
On-chain	Hashed product key, metadata hash, manufacturer and current owner, ownership history, stage status and commitment hash, evidence hashes, events

Location	Content
SQLite	Users and password hashes, audit events, product and raw-material records, quality templates and results, specifications, issuer registry, source registry and profiles, approvals and transaction references
File system	Evidence bytes under backend/uploads, one directory per stage, random generated names

5.11 Methodology

The approach to the problem was iterative. Requirements were first captured in a software requirements specification and a database design document. A narrow demonstration was then built that preserved the key boundary (off-chain metadata, on-chain hash) and was extended in layers: authentication and role-based access, raw-material and finished-product registration, versioned specifications, quality templates and gates, evidence upload and hashing, issuer trust and governance, the standards registry, and finally consumer verification with redaction. Each layer was accompanied by automated tests, and a fast presentation mode was added that drives the same real gates rather than bypassing them. The result is evaluated against the parameters in chapter 6.

Chapter 6

Experimental Setup

6.1 Database and Input Data

The system operates on a local SQLite database and on seeded fictional data; no real personal, commercial or laboratory data are used. The inputs are summarised in Table 6.1. Seeding is deterministic and idempotent.

Table 6.1: Seeded input data of the demonstration

Item	Details
Operational users	Five fictional accounts (supplier, manufacturer, distributor, retailer, admin), Argon2-hashed passwords
Flagship product	Aurelia Prestige Renewal Serum, code OC-LUX-SERUM-0001, batch APR-2026-001 (fictional)
Quality templates	24 checks: 5 supplier, 8 manufacturer, 5 distributor, 6 retail
Product specification	One versioned specification with 16 requirements (identity, bottle and dropper, seal, label, carton, appearance)
Evidence issuers	Six fictional issuers with TRUSTED_DEMO, UNVERIFIED or REVOKED status
Standards registry	Metadata for the Cosmetics Rules 2020, selected BIS standards and ISO 22716, 17516, 11930 and 29621, plus visible draft successors
Evidence files	PDF, PNG or JPEG, maximum 5 MiB, uploaded during the workflow

6.2 Performance Evaluation Parameters

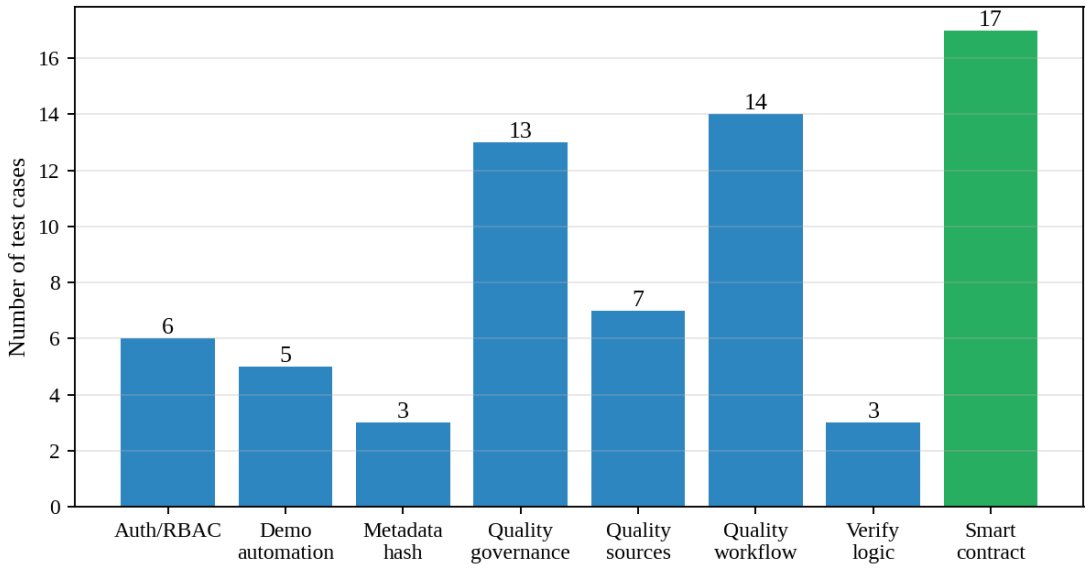
Because the project is a correctness-and-integrity demonstration rather than a throughput benchmark, validation uses the parameters in Table 6.2.

Table 6.2: Evaluation parameters used for validation

Parameter	How it is validated
Functional correctness	End-to-end workflow from raw material to APPROVED_FOR_SALE, including HOLD and correction
Gate enforcement	Transfers and approvals blocked while mandatory checks or evidence are missing
Tamper detection	Edited metadata produces SUSPICIOUS with a hash mismatch
Evidence integrity	VERIFIED, MISMATCH or FILE_MISSING when bytes are re-hashed
Issuer-trust handling	Revoked issuer cannot satisfy new critical evidence; history preserved
Access control	Role and stage rules enforced; admin cannot act as a supply-chain actor
Consumer redaction	No internal notes, paths, identifiers or internal evidence in public responses
Contract safety	Only the current owner can transfer or commit; duplicates and invalid inputs rejected
Test pass rate	Automated backend and contract test suites

Graph 6.1 shows the number of automated test cases per suite in the current build: 51 backend tests and 17 contract tests, 68 in total. The handover review of 5 October 2026 recorded all of

them as passing, together with the live API smoke test, the demo auto-fill flow, the revoked-issuer drill and a headless-browser interface smoke test.



Graph 6.1: Automated test cases per suite

6.3 Software and Hardware Setup

Table 6.3: Software requirements

Component	Version / tool
Operating system	macOS, Linux or Windows (start scripts provided)
Backend	Python 3.13 (3.10+ required); FastAPI 0.116+, Uvicorn, Web3.py 8, PyJWT, pwdlib with Argon2, qrcode, python-multipart
Blockchain	Node.js 22.13+, Hardhat 3.15, ethers 6, Solidity 0.8.28+
Database	SQLite
Frontend	HTML, CSS and JavaScript (no build step)
Testing	pytest 9, Hardhat Node test runner, headless Chrome smoke test

Table 6.4: Hardware requirements

Item	Specification
Processor	Any modern dual-core CPU or better
Memory	8 GB RAM recommended for running the node, API and browser together
Storage	About 1 GB free for dependencies, database and uploaded evidence
Network	None required at run time; all services run on localhost

Chapter 7

Project Plan

7.1 Gantt Chart

Figure 7.1 presents the project schedule. Solid bars are completed or in-progress work; hatched bars are planned future work.

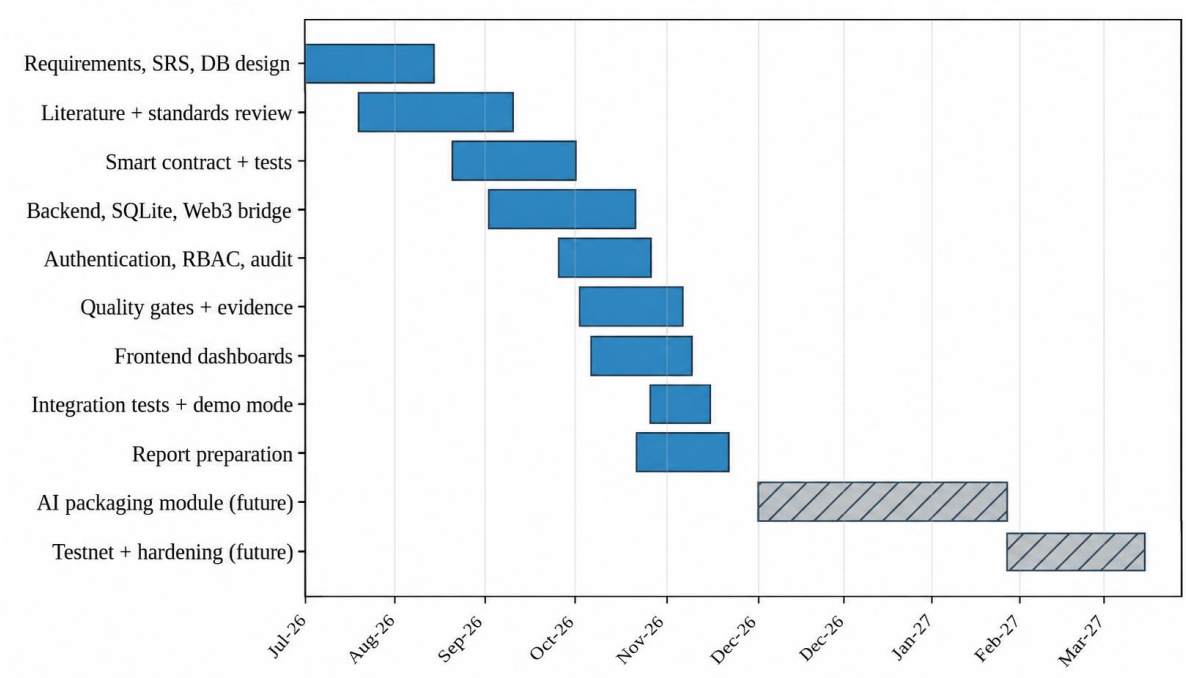


Figure 7.1: Gantt chart of the project plan

7.2 Team and Responsibilities

Table 7.1: Team roles

Member	Role
Durva Waghchaure	Backend Lead and Project Lead
Aradhya	Frontend Lead
Aryan	Blockchain Lead
Purvika	AI Lead (future packaging-inspection research)

Phases ran roughly in the order shown: requirements and design, literature and standards review, smart contract, backend and authentication, quality gates and evidence, frontend, integration and demo automation, and report preparation. The activities after the report (an AI packaging module using a rights-cleared labelled dataset, and testnet deployment with production hardening) are planned future work.

Chapter 8

Expected Outcome

From the analysis in the implementation chapters, the following outcomes are expected and are stated separately.

1. A working end-to-end system in which a fictional luxury serum moves from raw-material supplier to retailer only after each stage passes its mandatory checks with valid evidence.
2. Tamper-evident product identity: any edit to a stored product record is flagged as Suspicious because its hash no longer matches the on-chain value.
3. Verifiable evidence: every approval is tied to hashed evidence whose integrity can be re-checked, with VERIFIED, MISMATCH or FILE_MISSING reported.
4. Auditable operations: each meaningful action is attributed to an authenticated user in an admin-only audit log.
5. Layered access control: server-side role and stage rules plus independent owner checks in the contract.
6. A privacy-preserving consumer view that shows authenticity, the stage journey and public evidence without internal data.
7. Traceable quality requirements mapped to versioned standards metadata, with no certification or compliance claim.
8. A tested, reproducible codebase (51 backend and 17 contract tests) that can be extended to further product categories, a public testnet and an evidence-based AI module.

Chapter 9

Projected Cost Analysis

The demonstration itself uses open-source software and a local blockchain, so its incremental cost is close to zero. Table 9.1 gives indicative estimates for Project cost.

Table 9.1: Projected cost (indicative estimates)

Component	Technology Used	Estimated Cost
Smart Contract and Local Blockchain	Solidity, Hardhat	0
Backend API	Python, FastAPI, Web3.py	0
Database and Evidence Files	SQLite, Local Disk	0
Frontend	HTML, CSS, JavaScript	0
Blockchain Transaction Fees	Hardhat Test ETH	0
Hosting	Developer Laptop	0

The current prototype has zero cost, as it uses only open-source tools and runs locally. I measured the project's actual blockchain usage: each product batch needs 14 transactions. On the Polygon network that comes to about $0.03 per batch, compared with up to $25 on Ethereum mainnet. A small hosted pilot of 100 batches a month would cost about $15 per month, mostly for the server.

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